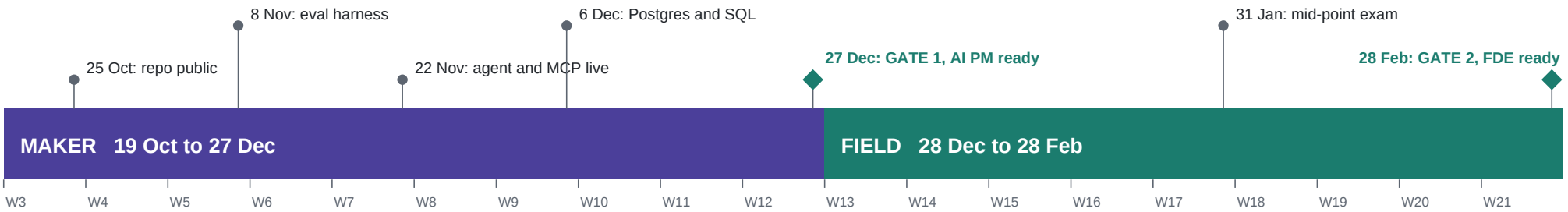
FORGE at a glance

Three phases, one hard stop. 19 Oct 2026 to 28 Feb 2027, then revision only.



49 hours a week (36 technical)

52 hours a week (39 technical). Diwali and New Year are normal weeks.

From 1 Mar: SHARPEN, revision only (no new topics). Gate 2 on 28 Feb closes FIELD.

The five projects and the platform

1 Call insights copilot: refund and disclaimer checks, agents, MCP, evals. 2 Contact-center data: warehouse, quality, mining, ML. 3 Customer integration service: login, webhooks, queues, deploys, telephony connector.

4 Collections recovery assistant: ranking, checked messages, payment plans, fairness. 5 Algo trading lab: backtest, risk limits, paper trading only. FORGE platform: the portfolio site with login and spend caps.

The rule that holds everything together

Track 1 (AI PM) is ready by 27 Dec. Track 2 (FDE, Staff TPM) is ready by 28 Feb. Every topic is taught with examples, exercises, flashcards and a link to one of your real projects. Nothing is added after 28 Feb except revision.

How every feature gets built

The same seven steps for every module in every project. Claude Code and Cursor each have a clear job.

1 Spec

One page: goal, inputs, outputs and failure cases.

Owner: you.

2 Tests first

Write tests for the failure cases before any code.

Owner: you, the agent can draft them.

3 Build

Claude Code for multi-file work, with a stop rule in the brief.

Cursor for small edits you want to watch.

Model: Haiku 5.5 only.

4 Review

Read every diff yourself.

Start a fresh session to review the work.

5 Verify

Run the tests and the evals yourself.

CI blocks a merge if they fail.

6 Commit

Save on a branch. One agent per branch.

Write a decision note for each real choice.

7 Explain

Explain the feature aloud in 2 minutes.

Add flashcards and a link to a career story.

Rules that never change

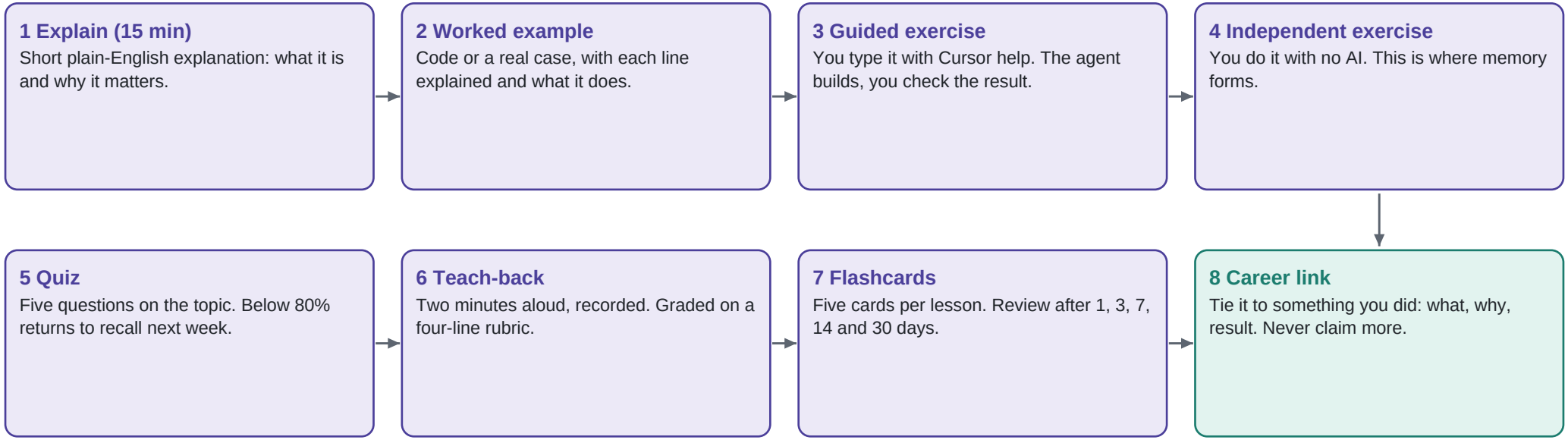
Haiku 5.5 is the only model, in Cursor and in Claude Code. Synthetic data only. No customer or Uniphore data in any prompt.

API keys live in a .env file that is never committed. Maximum 52 hours a week.

Cursor (built-in models) for edits. Claude Code CLI in a separate terminal for builds. Every brief has a stop rule, for example: stop after 3 failed tries.

How each topic is learned

Eight steps per topic, then the assessment rhythm that stops it from fading.



Then the next topic starts at step 1. The loop repeats for every topic from Week 3 to Week 21.

Rhythm	What	Time
Daily	Flashcards and three mixed questions	5 to 10 min
Twice a week	Blank-page recall: write what you remember, then check	15 min
Weekly	No-AI rebuild of one core function, plus a Sunday mock screen (10 questions)	30 min + 15 min
Checkpoints	8 Nov, 22 Nov and 6 Dec: 20 questions each	30 min
FIELD, every 2 weeks	Timed build with no AI, then explain it	45 min
FIELD, monthly	Mock loop: design drill, debug lab, behavioral	90 min
Exams	27 Dec, 31 Jan and 28 Feb: 60 questions plus a 3-hour prototype. Pass at 80%.	Half day each
SHARPEN (March on)	Weekly spaced recall. Monthly 20 mixed questions. Quarterly full exam (Jun, Sep, Dec 2027).	30 min a week

Why this order

Each step needs the one before it. The reason for each step is printed under it.

MAKER (19 Oct to 27 Dec): AI PM ready

1 Basics
Why: every later step calls code and a model. Gate, Pydantic and pytest come first.

→

2 RAG
Why: answers need sources before anyone can trust them.

→

3 Evals
Why: you cannot improve what you cannot measure. Built before agents.

→

4 Agents, MCP
Why: tools turn answers into actions, so they need evals and guardrails first.

→

5 Data and SQL
Why: product claims need numbers: outcomes, funnels and A/B tests.

→

6 Deploy
Why: customers run it, not just watch a demo. Webhooks, Docker and CI.

→

GATE 27 Dec
Track 1 ready: exam at 80%, public repo, PRD, six rehearsed stories, three mock loops.

FIELD (28 Dec to 28 Feb): FDE ready

7 Collections
Why: a new domain on the same AI core. Telephony brings real calls into the copilot.

→

8 AI depth
Why: hybrid search, tracing and an LLM gateway: skills that test well in interviews.

→

9 Cloud
Why: FDEs deploy into customer environments: Terraform, Kubernetes, backup and restore.

→

10 Integrate
Why: five projects and the platform joined by events. Shows real systems working together.

→

11 Interviews
Why: mock FDE and TPM loops. Practice under time, not only reading.

→

12 Final exam
Why: a prototype and a design drill that prove readiness.

→

GATE 28 Feb
FDE ready: five demos, a two-account deploy, CI with evals, exam at 80%.

Stretch items (Kubernetes, Terraform, Airflow, Hugging Face, LangChain, Kafka, voice latency) stay in full. They sit in FIELD, so MAKER's core is never at risk.

MAKER: Weeks 3 to 12

19 Oct to 27 Dec 2026. 49 hours a week (36 technical). Weeks 1 and 2 are done.

Week	Dates	Focus (technical first, then product)	Milestone
3	19 to 25 Oct	Python gate, structured output, eval v0 with precision and recall. Spec, tests, build, review loop. Secrets check. CV v1 (6 h). Metric tree, RICE, intro pitch.	25 Oct: repo public, eval v0
4	26 Oct to 1 Nov	RAG: chunks, embeddings, local index, recall at k and MRR. Data extraction from CSV and REST. Customer conversation 1. PRD v0.	RAG working with citations
5	2 to 8 Nov	Evals depth: 80-case golden set, LLM judge checked against 20 human labels, regression runs, cost per call, Hindi and Arabic test cases. Pricing sheet.	8 Nov: eval harness, checkpoint 1
6	9 to 15 Nov	Agent loop with two tools, state and tracing. API keys, FastAPI and OpenAPI. Speech lab: word error rate, mono vs. stereo. Telephony basics (concept).	Agent loop v1
7	16 to 22 Nov	MCP server and client, mock CRM as a tool, webhooks with retries and dead-letter queue, Redis queue. Model change test. Data protection note (critical).	22 Nov: agent and MCP live, checkpoint 2
8	23 to 29 Nov	PII redactor, red team (injection in tool results), cost chart. Authorization and audit log. Repeat-caller model: rules vs. classic ML vs. LLM. AI regulation one-pager. PRD v1.	PRD v1 final
9	30 Nov to 6 Dec	Postgres with pgvector in Docker Compose. SQL window functions, EXPLAIN, indexes. A/B sample size. Batch loader, dbt, SFTP. Platform shell starts.	6 Dec: Postgres and SQL
10	7 to 13 Dec	FastAPI service, webhook with HMAC signature, Dockerfile, Compose, GitHub Actions running pytest and evals. Dependency scan. Architecture document.	13 Dec: fallback decision, checkpoint 3
11	14 to 20 Dec	AWS deploy (billing alert first), second-account deploy, runbook, clean-machine test, observability, demo UI. Interview-mode rule starts.	Service live in AWS
12	21 to 27 Dec	Collections thin version. Timed prototype 1, incident simulation, value case, mock FDE. Fix the gaps from the checkpoints.	27 Dec: GATE 1, Phase 1 exam

FIELD: Weeks 13 to 21

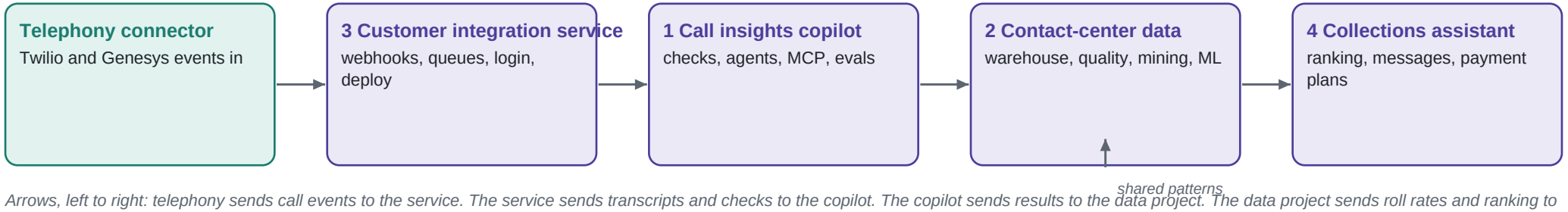
28 Dec 2026 to 28 Feb 2027. 52 hours a week (39 technical). Hard stop on 28 Feb.

Week	Dates	Focus (technical first, then product)	Milestone
13	28 Dec to 3 Jan	Collections build 1: gradient boosting, calibration, SHAP. Trees, cross-validation, ROC-AUC, imbalance, leakage. MLflow. Discovery questions and stakeholder map.	Propensity model baseline
14	4 to 10 Jan	Collections build 2: call assist, payment plan agent with hard limits, hardship routing. Redis at scale, outbox pattern. Telephony lab: Twilio and Genesys Cloud sandbox (8 h). Amazon Connect explain and try.	Telephony events reach the service
15	11 to 17 Jan	Collections 3 and data: recovery dashboard, roll rates in dbt, A/B test of message styles, fairness report, drift. Airflow or Prefect. Lineage, PII tagging, retention job. SMS reminders through a Twilio sandbox.	Collections dashboard live
16	18 to 24 Jan	AI depth: hybrid search and reranking, prompt caching and batch mode, LLM tracing, read-only text-to-SQL, LLM-assisted labeling. Hugging Face classifier. Local model. Trading start: price data, walk-forward backtest.	Tracing on the copilot
17	25 to 31 Jan	Agents and voice: permissions, context engineering, LangChain or LangGraph rebuild, voice latency (concept). TypeScript client. Load test with Locust or k6. OpenAPI contract-first.	31 Jan: mid-point exam
18	1 to 7 Feb	Cloud and infra: Terraform, cloud networking, secrets rotation, backup and restore test. Local Kubernetes. Threat modeling. Trading risk limits in code and LLM trade journal.	Restore test passed
19	8 to 14 Feb	Integrate and harden: all projects joined by events, end-to-end demo, model change test on all, OWASP LLM checklist. Dashboards, alerts, DORA. Telephony connector page on the platform (2 h). Kafka basics (explain). Paper-trading run.	End-to-end demo; decide on the release valve
20	15 to 21 Feb	Interview mastery: READMEs and demo videos, decision records, Staff TPM program plan. CV v2 for FDE roles (4 h). Mock loops for FDE, AI PM and Staff TPM. Platform landing and project pages.	Five demos ready
21	22 to 28 Feb	Final assessment: closing retros and gap list, final 3-hour prototype and design drill, final exam. Portfolio check: five repos, five demos, the live platform with spend caps on.	28 Feb: GATE 2, FDE ready

How the projects connect

One platform fronts all five projects. Telephony feeds the service and the copilot.

FORGE platform (portfolio site and SaaS demo)
Managed login, per-user API keys, usage metering, rate limits, spend caps, kill switch, one page per project.



Arrows, left to right: telephony sends call events to the service. The service sends transcripts and checks to the copilot. The copilot sends results to the data project. The data project sends roll rates and ranking to collections.

5 Algo trading lab
Own pipeline, paper trading only. Reuses the data patterns.

AI core in every project
LLM calls, structured output, RAG, evals, agents, MCP, guardrails, PII redaction, cost tracking, model change tests.

Track 2 strands under all of it
Backend and integration, data, classic ML, design and FDE skills. Each strand is tied to a project, so nothing is built twice.

Telephony add-on: Genesys and Twilio

About 9 extra technical hours. Your Genesys experience at Uniphore makes this a strong story.

Week 6: concept (1 h)

How a call travels: phone network, SIP, IVR menus, DTMF key presses, call recording, streaming audio.

Explain and try. Career link: the U-Assist Genesys telephony delay and the phased release.

Week 14: lab (8 h, build)

Twilio: voice webhook, TwiML reply, status callbacks, signature check.

Genesys Cloud developer sandbox: an Architect flow that calls a data action on your service.

Result: a call-ended event reaches the service and the copilot.

Week 15: SMS (in collections)

Reminder SMS through a Twilio sandbox, checked by the message checker.

India: sender and template registration (DLT) is needed for real sends. Learn the rule; use the sandbox only.

Week 19: connector page (2 h)

Telephony connector on the FORGE platform: connect a provider, see call events, see the checked result of a recording.

Amazon Connect stays explain and try, if its trial is free.

Item	Cheapest route	Check first
Twilio	Free trial credit. Verified test numbers only.	Current trial limits, and whether a number is included
Genesys Cloud	Developer or trial org, if offered.	Current developer terms and API limits
Phone numbers	Do not buy one for learning. Test webhooks through a free tunnel tool.	Monthly charge, if you ever buy a number
SMS in India	Sandbox only for learning.	DLT registration rules before any real send
Call recording	Record only test calls with your own consent.	The consent rule for each market (India, Gulf, Philippines)
Usage costs	Pay per call or SMS, in small amounts.	Set a spend limit on each account

Hours and the release valve

Option B, as frozen. Rough estimates, shown so you can see the margin.

Phase	Hours a week	Technical hours	Notes
MAKER (Weeks 3 to 12)	49 (36 technical)	360	No lighter Diwali or New Year weeks
FIELD (Weeks 13 to 21)	52 (39 technical)	351	All stretch items included
Total technical hours		711	Option B

What the full scope needs (rough)

- Original scope: about 650 to 700 technical hours.
- Platform and trading project: about 33 more. Telephony: about 9 more.
- Total: about 690 to 740 hours. Option B gives 711, so the margin is between about 30 hours short and about 20 hours spare.

The release valve, decided once

If FIELD runs short, move the end date to 15 March. That adds two weeks, about 78 technical hours.

Decide at the start of Week 19 (8 Feb), not before. Until then, keep Option B and protect the gates: the 27 Dec and 28 Feb gates are never cut.

Not in this PDF: the learning-usage ceiling (still to be set) and the platform ceiling (INR 1,000 a month, excluding Cursor Start).